A CASE-STUDY BASED REPORT

ON

Accident Hotspot Prediction Using Random Forest and Spatial Clustering

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

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Abstract

Road accidents are a serious public-safety problem in fast-growing cities, and the agencies that manage road safety have to decide which locations deserve attention first. This case study applies machine learning to the official list of accident black spots identified for Hyderabad City, which records 49 accident-prone road stretches across seven police zones together with the number of accidents and fatalities in 2023, 2024 and 2025 (398 accidents and 279 fatalities in total).

The raw CSV file, which carries several banner and multi-level header rows, was loaded and cleaned with Python and pandas. Seven additional features were engineered, including average yearly accidents, accident growth, fatality rate and a severity score defined as Total Accidents + 1.5 × Total Fatalities. Using documented thresholds on this score, each location was labelled LOW, MEDIUM or HIGH risk (20, 17 and 12 locations respectively). A Random Forest classifier with 100 trees and a maximum depth of 5, trained on a stratified 70/30 split, reached 73.33% accuracy and a weighted F1 score of 72.81% on the 15 held-out locations.

Because the dataset contains no latitude or longitude values, true geographic clustering was not possible and no coordinates were invented. Instead, K-Means clustering (k = 3) on standardised severity features was used as a feature-space proxy, which separated five critical blackspots from 21 high-severity and 23 moderate-severity locations. Exploratory analysis showed that accidents fell from 150 in 2023 to 113 in 2025, that the Shamshabad zone has the highest average severity, and that 26 of the 49 locations lie on national highways. The work is delivered as a menu-driven Python terminal application that can reproduce every table, chart and prediction reported here.

Keywords: accident black spots, Hyderabad, Random Forest, K-Means clustering, severity score, exploratory data analysis, road safety.

Introduction

Background

Road traffic crashes cause a large number of deaths and injuries every year, and India is among the countries with the highest road-traffic fatalities in the world. Accidents are not spread evenly over a road network. They tend to concentrate at particular junctions, curves, bridges, highway stretches and market areas. Such concentrations are called accident black spots, and road-safety authorities identify them by counting the accidents and fatalities that occur on a defined stretch of road over a period of several years.

This project works with the official statement of black spots identified for Hyderabad City for calendar year 2025. For every black spot the statement gives the zone, the jurisdictional police station, the national-highway or other road on which it lies, the name and limits of the stretch, and the number of accidents and fatalities in each of the years 2023, 2024 and 2025. The data are small but they are real, which makes them well suited to a case study that is meant to be understood and defended end to end.

Motivation for the Project

Engineering fixes such as speed breakers, signage, pedestrian crossings, junction redesign and enforcement are expensive, and no city can apply them everywhere at once. A transparent, data-driven way of ranking and grouping black spots helps decision makers justify why one location is treated before another. The team was also motivated by the chance to apply the full machine-learning workflow taught in the Machine Learning course (data loading, cleaning, feature engineering, exploratory analysis, supervised learning, unsupervised learning and evaluation) to a problem that affects people in the city where the data were collected.

Overview of the Work Carried Out

The work was implemented as a modular Python application with a numbered terminal menu. The main stages were:

- **Data loading:** reading the raw CSV, skipping the four title and header rows and assigning 15 standard column names.
- **Preprocessing and feature engineering:** cleaning text fields, converting data types, handling missing values, verifying totals and creating seven derived features and the target risk label.

- **Exploratory data analysis:** five summary tables and four charts covering yearly trends, top locations, zones and the risk distribution.
- **Random Forest classification:** training, evaluation (accuracy, precision, recall, F1 score, confusion matrix) and feature-importance analysis.
- **Spatial clustering and hotspot ranking:** a coordinate audit, K-Means feature-space clustering and a ranked list of hotspots by severity.
- **Prediction and reporting:** an interactive risk predictor for an existing or user-defined location and an automatic text report generator.

Scope and Responsible-Use Statement

Scope. The study covers the 49 black spots of Hyderabad City listed in the supplied dataset and the three calendar years 2023 to 2025. The data are aggregated counts per road stretch. The study does not use individual crash records, vehicle types, causes, time of day, weather or traffic volume, and it does not forecast accidents in future years. Its output is a classification of existing black spots into relative risk levels and a grouping of similar locations.

Responsible use. The dataset contains no personal information about victims or drivers. The risk levels and rankings produced here are analytical results derived from one dataset and one team-defined scoring rule. They are not an official government ranking or statutory designation, and the application prints a disclaimer to that effect. They should be used to support, and never to replace, the judgement of traffic engineers and enforcement authorities. Because the dataset has no geographic coordinates, none were fabricated, and the clustering results are explicitly described as a proxy rather than a geographic analysis.

Problem Statement

Case-Study Scenario

Consider the road-safety wing of a city traffic authority that holds the official black-spot statement for Hyderabad City. The statement lists 49 locations but gives only raw counts. The authority has a limited budget for the coming year and must decide which stretches should receive speed monitoring, pedestrian overpasses or junction improvements first. It also wants to know whether the locations fall into a small number of recognisable profiles so that one type of intervention can be planned for each profile, and whether a simple model can assign a risk level to a location from its accident history.

Central Case-Study Question

Can the historical accident and fatality counts of Hyderabad City's black spots be used to classify each location as LOW, MEDIUM or HIGH risk and to group similar hotspots, so that safety resources can be directed first to the most dangerous locations?

Formal Problem Definition

Let D be the set of n = 49 black spots. For each location i, let A(i) be the total accidents and F(i) the total fatalities over 2023 to 2025, and let x(i) be a feature vector of 17 values (ten numeric features and seven one-hot zone indicators). The severity score and the target label are defined as follows.

```
Severity Score S(i) = A(i) + 1.5 x F(i)

Risk label y(i) = LOW     if S(i) <= 13.0
                  MEDIUM  if 13.0 < S(i) <= 18.0
                  HIGH    if S(i) > 18.0
```

The problem has three parts: (1) a supervised task, to learn a function f that maps x(i) to y(i) using 70% of the locations and to measure accuracy, precision, recall and F1 score on the remaining 30%; (2) an unsupervised task, to partition the locations into k = 3 groups using the standardised values of total accidents, total fatalities, severity score and average yearly accidents; and (3) a ranking task, to order all locations by severity score and report the most dangerous hotspots.

Objectives that Guide the Study

1. Load and clean the multi-row-header black-spot dataset reliably and document every preparation step.

2. Engineer meaningful risk indicators (average yearly accidents, accident growth, fatality rate, severity score and stretch length).
3. Explore the data to understand yearly trends, zone-wise concentration, highway involvement and the distribution of risk.
4. Train and evaluate a Random Forest classifier to predict the risk level of a location and identify the factors that influence it.
5. Group locations into risk profiles with K-Means clustering and rank the highest-severity hotspots.
6. Provide an interactive application and an exportable report so that the analysis can be reproduced and explained.

Assumptions and Limitations

Assumptions

- The official counts of accidents and fatalities are accurate and comparable across the three years.
- A blank fatality entry means no fatality was recorded and is treated as 0. This affected four cells in two locations.
- A fatality is more serious than a non-fatal accident, so it is weighted 1.5 times in the severity score. The weight and the cut-offs of 13.0 and 18.0 are fixed by the team in the code (the upper cut-off coincides with the 75th percentile of the severity score).
- Each location is treated as an independent record.

Limitations

- **Small sample.** With 49 records the training set has 34 and the test set only 15 locations, so every metric is sensitive to which locations fall in the test set.
- **Label derived from the inputs.** The risk label is calculated from total accidents and total fatalities, and the yearly counts that sum to those totals are also model inputs. The classifier is therefore largely re-learning a known rule rather than predicting an independent outcome, and its accuracy should be read in that light.
- **No coordinates.** The dataset has no latitude or longitude, so geographic clustering such as DBSCAN on GPS points was not possible. K-Means on severity features groups locations by risk profile, not by physical closeness.
- **Missing information.** Start and end kilometre values are blank for 22 locations (stretch length is set to 0 for these), and no data on traffic volume, road geometry, time of day or weather are available.

- **Aggregated, three-year data.** The counts cannot show causes of accidents or trends beyond 2023 to 2025.

Dataset and Exploratory Data Analysis

About the Dataset

The dataset is the statement titled "Black spots identified by the State in calendar year 2025 in respect of Hyderabad City", supplied as a CSV file (AccidentProneAreas_Complete). The file starts with a title row, a district row and two levels of column headers, so the actual records begin on the fifth row. After loading, it contains 49 records and 15 columns. The records are spread over seven zones (Secunderabad, Rajendranagar, Golconda, Jubilee Hills, Shamshabad, Charminar and Khairathabad) and 24 jurisdictional police stations. Of the 49 stretches, 26 lie on national highways (NH-44, NH-65, NH-163 and NH-765) and 23 are on other roads.

Table 1: Columns of the dataset after loading

Column	Description	Type
sl_no	Serial number of the black spot	Integer
zone	Police zone of Hyderabad City	Text
police_station	Jurisdictional police station	Text
road_info	National highway or other road on which the spot lies	Text
location_name	Name and limits of the accident-prone stretch	Text
start_km, end_km	Starting and ending kilometre of the stretch	Decimal
accidents_2023, 2024, 2025	Number of accidents in each year	Integer
total_accidents	Total accidents over the three years	Integer
fatalities_2023, 2024, 2025	Number of fatalities in each year	Integer
total_fatalities	Total fatalities over the three years	Integer

Purpose of EDA

Exploratory data analysis was carried out before any model was built for four reasons. First, to check the quality of the data and find missing, inconsistent or duplicated values. Second, to understand how accidents and fatalities are distributed across years, zones, roads and locations. Third, to confirm that the engineered severity score and the three risk classes give a sensible, reasonably balanced target. Fourth, to present the data visually in a way that a non-technical reader, such as a traffic official, can follow.

Preparation Steps Actually Performed

The following steps are implemented in data_loader.py and preprocessing.py and were all executed on the supplied file.

1. **Loading.** The CSV was read with the first four rows skipped and no header, the first 15 columns were given standard names, and completely blank rows were dropped. This produced 49 records.
2. **Text cleaning.** Leading and trailing spaces were removed from the zone, police station, road and location columns, and the police-station spelling "OU Sity" was standardised to "O U Sity". After this step there are 24 distinct police stations.
3. **Type conversion and missing values.** All numeric columns were converted with pd.to_numeric. Missing values were set to 0: the 22 blank start/end kilometre values and four blank fatality cells in two locations (Thondupally and KBR Park).
4. **Duplicate check.** Duplicate rows were searched for and none were found (0 removed).
5. **Consistency check.** Total accidents and total fatalities were recomputed from the three yearly columns. They matched the totals given in the file for all 49 records (398 accidents and 279 fatalities).
6. **Feature engineering.** Seven new features were created, as listed in Table 2.
7. **Target creation.** The risk level was assigned from the severity score using the thresholds 13.0 and 18.0, giving 20 LOW, 17 MEDIUM and 12 HIGH locations.
8. **Model matrix.** The feature matrix X was built from ten numeric features plus the one-hot encoded zone (seven columns), giving 17 input columns. The risk level itself is excluded from X.

Table 2: Engineered features

Feature	Formula	Meaning
average_yearly_accidents	total_accidents / 3	Typical accidents per year
accident_growth	accidents_2025 - accidents_2023	Rise or fall over the period
fatality_rate	total_fatalities / total_accidents	Fatalities per recorded accident
severity_score	total_accidents + 1.5 x total_fatalities	Weighted seriousness of the spot
stretch_length_km	|end_km - start_km|	Length of the black-spot stretch
risk_level	Threshold rule on severity_score	Target label: LOW, MEDIUM, HIGH

The key preprocessing code is shown below.

```
df['total_accidents'] = (df['accidents_2023'] + df['accidents_2024']
                         + df['accidents_2025'])
df['severity_score'] = df['total_accidents'] + 1.5 * df['total_fatalities']
df['fatality_rate'] = (df['total_fatalities']
                       / np.maximum(df['total_accidents'], 1.0))

def assign_risk_level(sev):
    if sev <= 13.0:   return 'LOW'
    elif sev <= 18.0: return 'MEDIUM'
    else:             return 'HIGH'
```

EDA Exploration - Implementation

The EDA module (eda.py) prints five formatted tables in the terminal and saves four charts to the outputs/charts folder using pandas, matplotlib and seaborn. The core of the zone-wise analysis is a single group-by:

```
zone_grp = df.groupby('zone').agg(
    hotspots=('sl_no', 'count'),
    total_acc=('total_accidents', 'sum'),
    total_fat=('total_fatalities', 'sum'),
    avg_acc=('total_accidents', 'mean'),
    avg_fat=('total_fatalities', 'mean')
).reset_index().sort_values(by='total_acc', ascending=False)
```

Yearly trend

Table 3: Yearly accident and fatality summary

Year	Accidents	Fatalities	Fatalities per accident
2023	150	112	74.67%
2024	135	90	66.67%
2025	113	77	68.14%
Total	**398**	**279**	**70.10%**

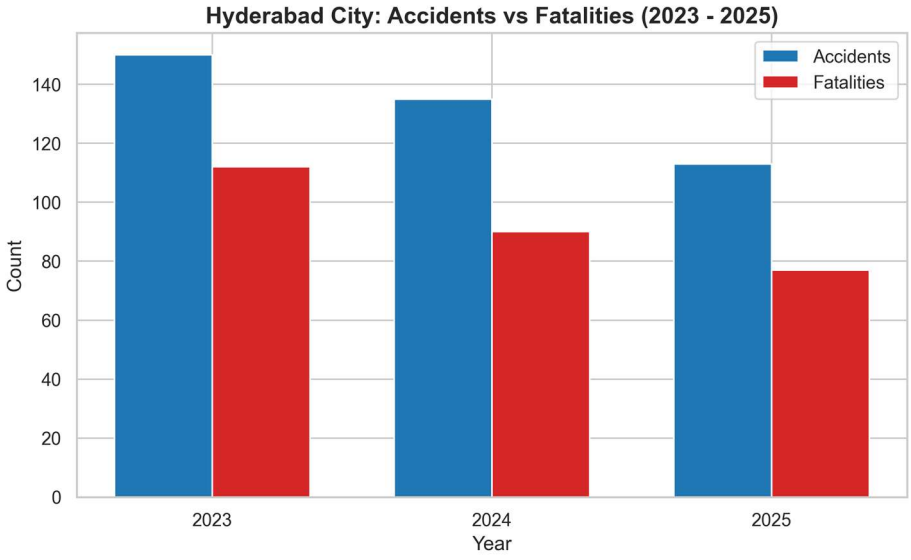


Figure 1: Accidents versus fatalities at Hyderabad black spots, 2023 to 2025

Accidents fell every year, from 150 to 135 to 113, a reduction of 24.7% between 2023 and 2025. Fatalities fell faster, from 112 to 77 (31.3%). The number of fatalities per recorded accident stayed high at about 0.67 to 0.75 in every year.

Top locations

Table 4: Ten locations with the most accidents (2023 to 2025)

Rank	Location	Zone	Acc.	Fat.	Risk
1	Shamshabad (MRO Office Kaman to Begum's India Garden Function Hall)	Shamshabad	21	21	HIGH
2	Bandlaguda X Road	Rajendranagar	17	13	HIGH
3	Habsiguda X Road	Secunderabad	17	12	HIGH
4	Gaganpahad	Shamshabad	15	15	HIGH
5	Tarnaka X Road	Secunderabad	15	10	HIGH
6	ESI Hospital / Erragadda	Jubilee Hills	14	6	HIGH
7	Thondupally	Shamshabad	13	0*	LOW
8	Malakpet Gunj / Chermas	Charminar	11	10	HIGH
9	Zoo Park / Tadbun X Roads	Rajendranagar	10	10	HIGH
10	Jubilee Hills Check Post	Jubilee Hills	10	2	LOW

**The fatality columns for Thondupally are blank in the source file and were treated as 0, which is why a location with 13 accidents is labelled LOW.*

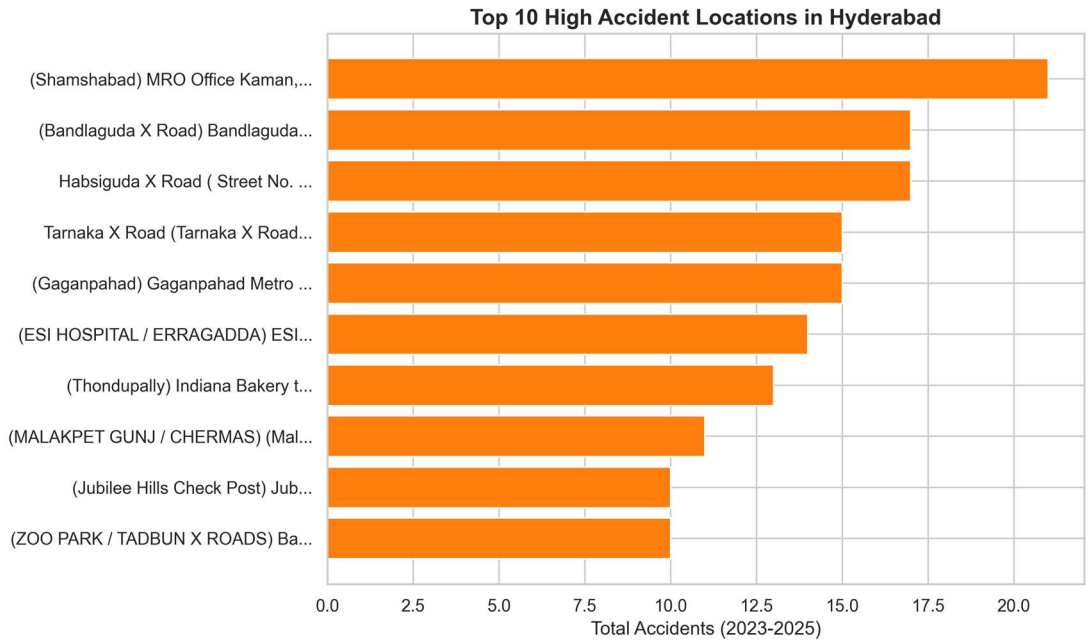


Figure 2: Top 10 high-accident locations in Hyderabad

Zone-wise distribution

Table 5: Accidents and fatalities by zone

Zone	Spots	Accidents	Fatalities	Avg acc. per spot	Avg fat. per spot	HIGH-risk spots
Secunderabad	10	85	55	8.50	5.50	2
Rajendranagar	10	84	69	8.40	6.90	4
Shamshabad	5	63	48	12.60	9.60	3
Golconda	9	58	40	6.44	4.44	1
Jubilee Hills	8	58	28	7.25	3.50	1
Charminar	5	37	31	7.40	6.20	1
Khairathabad	2	13	8	6.50	4.00	0

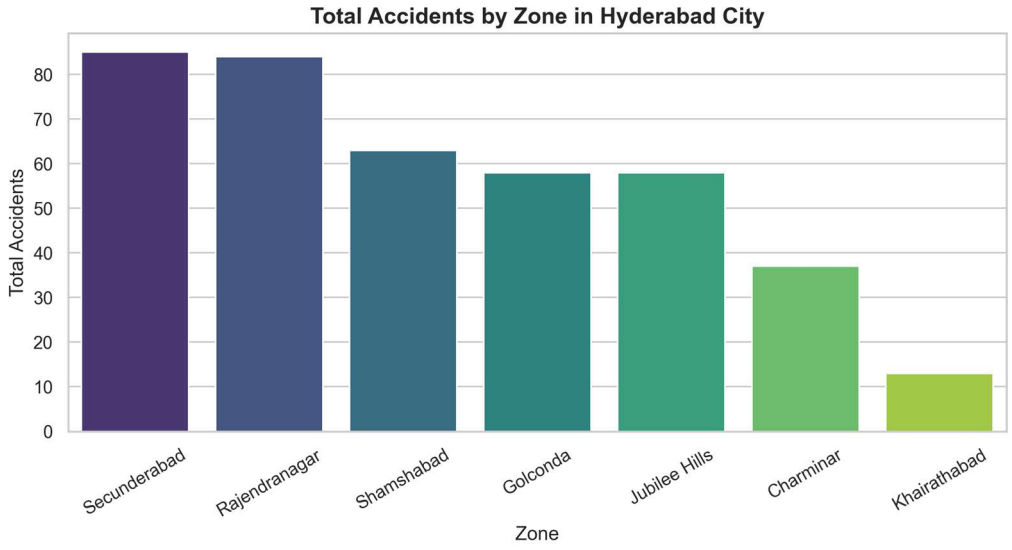


Figure 3: Total accidents by zone

Risk distribution

Table 6: Distribution of risk categories

Risk category	Severity score range	Locations	Percentage
LOW	S <= 13.0	20	40.8%
MEDIUM	13.0 < S <= 18.0	17	34.7%
HIGH	S > 18.0	12	24.5%

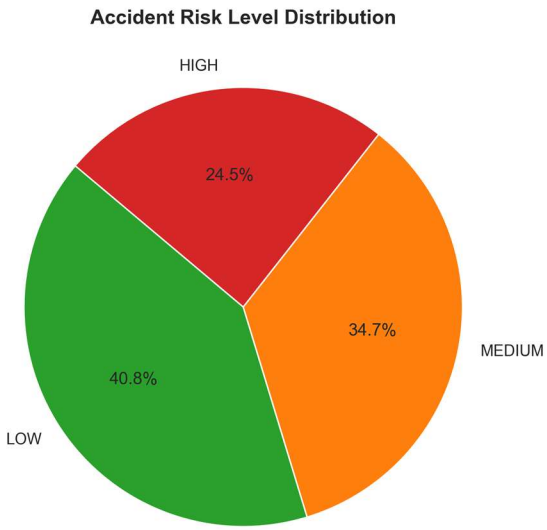


Figure 4: Accident risk level distribution

Summary of EDA Findings

1. **Declining trend.** Accidents at the 49 black spots dropped from 150 (2023) to 113 (2025), and fatalities from 112 to 77. At the location level, however, 15 spots saw more accidents in 2025 than in 2023, 26 saw fewer and 8 were unchanged. The largest increase was at ESI Hospital / Erragadda (+8) and the largest decrease at Pillar No. 100 to 116, Mogal ka Naala (-7).
2. **High lethality.** Across all spots there were 279 fatalities for 398 accidents (0.70 per accident). Only one location recorded zero fatalities. Total accidents and total fatalities are strongly correlated (correlation coefficient 0.76).
3. **Concentration in a few locations.** The average spot has about 8 accidents (median 7), but the worst spot has 21 accidents and 21 fatalities. Shamshabad (MRO Office Kaman) has a severity score of 52.5, far above the mean of 16.7.
4. **Zone pattern.** Secunderabad (85) and Rajendranagar (84) have the most accidents in total, and Rajendranagar has the most fatalities (69). Shamshabad has only five spots but the highest average per spot (12.6 accidents and 9.6 fatalities) and the highest average severity score (27.0). Rajendranagar has the most HIGH-risk spots (4).
5. **Highways matter.** The 26 spots on national highways account for 213 accidents and 159 fatalities, compared with 185 accidents and 120 fatalities at the 23 spots on other roads.
6. **Reasonably balanced classes.** The severity-based labels give 40.8% LOW, 34.7% MEDIUM and 24.5% HIGH locations, which is suitable for classification, although the HIGH class is small.

7. **Data-quality notes.** The file had no duplicates and its totals were consistent. The main gaps were 22 blank kilometre ranges and four blank fatality cells, and the lack of GPS coordinates.

Algorithm Selection and Implementation

Why these algorithms

Random Forest (supervised classification). The task is to assign one of three ordered risk classes to a location described by a small table of numeric and one-hot features. A Random Forest builds many decision trees on random samples of the data and features and takes a majority vote. It was chosen because it works well on small tabular data without feature scaling, captures non-linear relationships and interactions, is far less prone to over-fitting than a single decision tree, handles the mix of numeric and one-hot features in this dataset, and provides feature importances that can be explained to non-technical users. Logistic regression would assume linear class boundaries, a single tree would be unstable with 34 training records, and neural networks or SVMs need more data and are harder to interpret.

K-Means (unsupervised clustering). To group hotspots by risk profile, K-Means with k = 3 was used on the standardised values of total accidents, total fatalities, severity score and average yearly accidents. K-Means is simple, fast and easy to interpret, and k = 3 matches the three risk levels. It is used here as a proxy for spatial clustering because the dataset has no latitude and longitude; with coordinates, a density-based method such as DBSCAN would be the preferred choice.

Implementation

The Random Forest model is trained in random_forest_model.py. Stratification keeps the class proportions equal in the training and test sets, and a fixed random_state of 42 makes the results reproducible.

```
X, y = prepare_ml_features(df)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)

rf_model = RandomForestClassifier(
    n_estimators=100, max_depth=5, random_state=42)
rf_model.fit(X_train, y_train)
y_pred = rf_model.predict(X_test)
```

Table 7: Model configuration

Item	Setting
Algorithm	RandomForestClassifier (scikit-learn)
Number of trees (n_estimators)	100
Maximum tree depth (max_depth)	5
Input features	17 (10 numeric + 7 one-hot zone columns)
Target	risk_level (LOW, MEDIUM, HIGH)
Train / test split	70% / 30% (34 / 15 records), stratified, random_state = 42
Test-set class counts	LOW 6, MEDIUM 5, HIGH 4

Clustering is performed in clustering.py. The features are standardised first so that no single measure dominates the distance calculation.

```
cluster_cols = ['total_accidents', 'total_fatalities',
                'severity_score', 'average_yearly_accidents']
X_scaled = StandardScaler().fit_transform(df[cluster_cols])
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df['cluster_id'] = kmeans.fit_predict(X_scaled)
```

The application also includes a prediction module (prediction.py), where the user can pick an existing location or type custom feature values, and a report module (report.py) that writes the complete project report to outputs/reports.

Results and Evaluation

Random Forest performance

Table 8: Overall performance on the 15-record test set

Metric	Value
Accuracy	0.7333 (73.33%)
Precision (weighted)	0.7333 (73.33%)
Recall (weighted)	0.7333 (73.33%)
F1 score (weighted)	0.7281 (72.81%)

Table 9: Class-wise performance

Class	Precision	Recall	F1 score	Test records
LOW	0.83	0.83	0.83	6
MEDIUM	0.67	0.80	0.73	5
HIGH	0.67	0.50	0.57	4

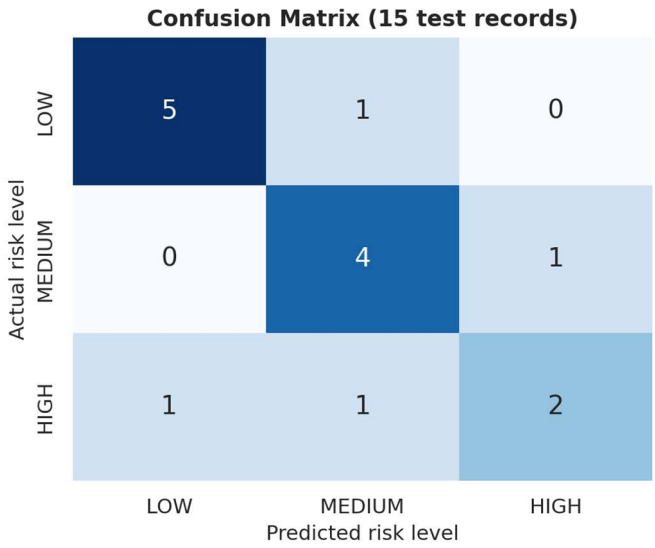


Figure 5: Confusion matrix on the test set

Eleven of the 15 test locations were classified correctly. The model is best at the LOW class (5 of 6 correct). The HIGH class is the weakest, with a recall of 0.50: of the four HIGH-risk locations in the test set, two were correctly identified, one was predicted MEDIUM and one LOW. All four errors are between neighbouring or nearby classes, and one MEDIUM location was predicted HIGH, which is the safer direction of error for a safety application.

Feature importance

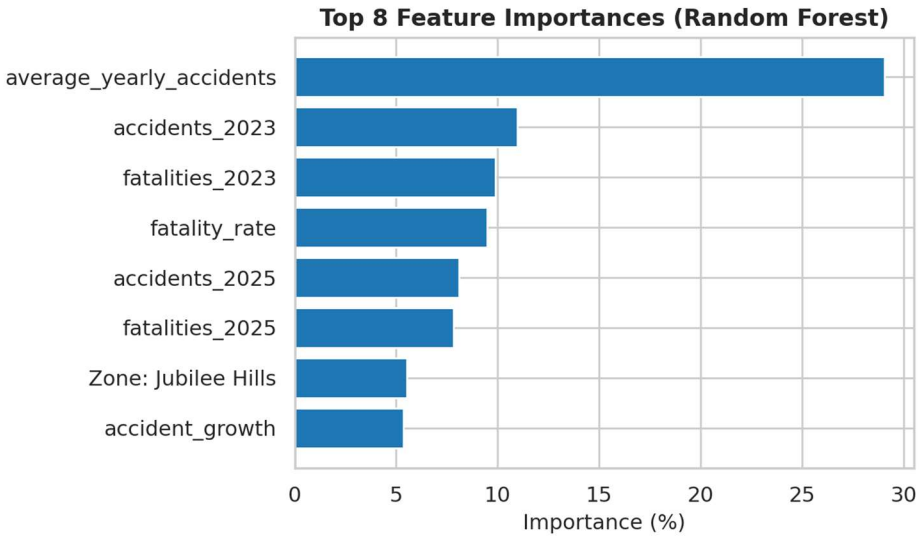


Figure 6: Top 8 feature importances of the Random Forest

Average yearly accidents is the most influential feature (29.07%), followed by accidents in 2023 (10.98%), fatalities in 2023 (9.89%), fatality rate (9.47%) and accidents in 2025 (8.08%).

Accident-count features dominate, which is expected because the target is built from accidents and fatalities. The zone indicator for Jubilee Hills is the most important location-based feature (5.52%).

Robustness check

Because the test set has only 15 records, the team also ran a supplementary 5-fold stratified cross-validation with the same model settings. The fold accuracies were 100%, 70%, 70%, 80% and 88.9%, with a mean of 81.8%. This check was run for this report and is not part of the application. The spread between folds confirms that a single 15-record split gives only a rough estimate, and the official project result remains the 73.33% reported above.

K-Means clustering results

Table 10: Cluster profiles (feature-space K-Means, k = 3)

Cluster (label used in the application)	Locations	Avg accidents	Avg fatalities
Cluster 1: Critical Blackspots (extreme severity)	5	17.0	14.2
Cluster 2: High Fatality / Highway Corridors	21	9.0	5.6
Cluster 3: Moderate Risk Urban Intersections	23	5.4	3.9

The cluster names are tags that the application assigns by ranking clusters on mean severity. They describe the severity level of each group and were not verified against road type, so Cluster 2 and Cluster 3 should not be read as a literal split between highways and urban junctions. Cluster 1 contains the five most severe locations, which also head the hotspot ranking below.

Hotspot ranking

Table 11: Top five hotspots by severity score

Rank	Location	Zone	Acc.	Fat.	Severity
1	Shamshabad (MRO Office Kaman)	Shamshabad	21	21	52.5
2	Gaganpahad	Shamshabad	15	15	37.5
3	Bandlaguda X Road	Rajendranagar	17	13	36.5
4	Habsiguda X Road	Secunderabad	17	12	35.0
5	Tarnaka X Road	Secunderabad	15	10	30.0

Conclusion

This case study showed that the official black-spot statement for Hyderabad City can be turned into a reproducible machine-learning workflow. The raw file was cleaned and checked, seven risk indicators were engineered, and every location was labelled LOW, MEDIUM or HIGH with a documented severity rule. A Random Forest classifier reached 73.33% accuracy and a 72.81% weighted F1 score on the held-out locations, and K-Means grouped the black spots into three severity profiles, with five critical blackspots standing apart from the rest.

The analysis answers the central question in a practical way. Accident counts and fatalities are enough to rank and group the locations: Shamshabad (MRO Office Kaman), Gaganpahad, Bandlaguda X Road, Habsiguda X Road and Tarnaka X Road have the highest severity, the Shamshabad zone has the highest average severity, and national-highway stretches account for most fatalities. Although accidents and fatalities declined from 2023 to 2025, about one in three spots (15 of 49) recorded more accidents in 2025 than in 2023.

Based on these findings, the team suggests that the five highest-severity hotspots be considered first for measures such as speed monitoring, pedestrian overpasses and improved junction layouts, and that the Shamshabad and Rajendranagar zones be examined as priority areas. These are suggestions drawn from a limited dataset and should be validated by traffic engineers on site.

The study has important limits. The dataset is small, the risk label is computed from quantities that the model also receives as inputs, there are no coordinates, and the clustering is a feature-space proxy rather than a geographic analysis. Future work could (1) add latitude and longitude so that DBSCAN or GIS-based hotspot mapping can be used, (2) add traffic volume, road geometry, time-of-day and weather data, (3) use an independent target such as accidents in the following year, (4) compare other classifiers with cross-validation and hyperparameter tuning, and (5) present the results in a web dashboard with maps for decision makers.

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